

Developing an IoT and ML-driven platform for fruit ripeness evaluation and spoilage detection: A case study on bananas

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ABSTRACT

Food waste is a significant global problem that demands immediate action to reduce it. This study presents a novel framework that merges Internet of Things (IoT) and machine learning (ML) technologies to detect fruit ripeness and spoilage, which is essential in minimizing losses in the cold chain process of fresh produce industry. The study employed temperature, humidity, and gas emission sensors along with an ESP32 microcontroller to establish a unique framework that achieved exceptional accuracy in predicting banana ripeness stages. This framework employed various machine learning algorithms to detect ripeness stages, with the CatBoost classifier exhibiting exceptional performance, demonstrating its dependability and effectiveness in assessing fruit quality. The benefits of this research extend beyond fruit ripeness detection and pave the way for future advancements in automating quality assessment in agricultural supply chains.

1. Introduction

Food waste is a pressing problem globally and in India, which presents significant environmental and socioeconomic challenges. In India, with its growing urbanization and population, approximately 1.3 billion tonnes of food is wasted annually. That amounts to the produce of approximately 28 % of the country's total agricultural land. Each Indian citizen generates 0.5 kg of organic waste daily. The *United Nations Sustainable Development Goal 12.3* aims to halve global food waste by 2030, acknowledging its detrimental impact on land, water, energy, and greenhouse gas emissions [1,2]. An estimated 40–50 % of the world's fruits and vegetables are lost or wasted throughout the supply chain, accounting for approximately half of all food waste [2]. This leads to a waste of resources, unnecessary greenhouse gas emissions, and substantial economic loss [3]. Therefore, reducing food waste in the supply chain is essential for both environmental and economic reasons. Various factors contribute to high losses in the fresh fruit and vegetable sector, including weather and the lack of alternative markets. Ludwig-Ohm et al., [4] investigated on strawberries, apples, lettuce, and carrots and found out that, on average, 25 % of vegetables, 15 % to 20 % of strawberries and 6 % to 16 % of apples are lost at the farm level. Most of the food waste is attributed to fruits, vegetables, and grains. But to reduce losses, precautionary measures like protected cultivation,

continuous cooling, improved packaging, and staff training need to be considered. Besides that, increased cooperation between growers, wholesalers, and retailers can enhance the marketable product share. This highlights the importance of the environment in maintaining freshness and safety. It is crucial to continuously monitor their condition to ensure their quality and nutritional value. As the conventional means of tracking the condition of produce in supply chains is outdated and lack digitalization, the rapid spoilage of these items necessitates careful storage and transportation to extend their shelf life and preserve their post-harvest quality [5]. The ripeness of fruits is a fundamental aspect of agricultural production and supply chain industry. It is a critical determinant of the quality and market value of fruits, impacting both producers and consumers. For farmers and producers, accurate and timely ripeness detection is essential for optimizing harvesting schedules, reducing post-harvest losses, and ensuring that fruits reach consumers in their prime condition. Consumers, on the other hand, rely on the ripeness of fruits to make informed purchasing decisions, seeking produce that meets their taste preferences and nutritional requirements [6]. Fruit ripeness assessment typically relies on visual inspection by skilled personnel; however, this method is subjective, time-consuming, and prone to errors due to external factors such as fatigue and bias.

Present study considered banana fruit as India is one of the largest producers of Banana. Its availability throughout the year made it

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people's choice. However, due to infrastructural issues in logistics and cold chain processes, there is large amount of spoilage in banana produce resulting losses to the farmers as well as producers [7–10]. The losses are mainly due to lapses in assessing banana ripeness. The conventional method of observing manually is not advisable, because even if the professionals involved, inconsistencies arise from individual variations in perception. Therefore, the authors of this article proposed an automated system for analysing banana ripeness using emerging technologies such as Internet of Things (IoT) and Machine learning algorithms.

The primary aim of this study is to present an IoT system integrated with a machine-learning technique for fruit ripeness detection and spoilage reduction in banana.

The following are the key contributions of this work:

1. This study addresses the pressing global issue of food security and contributes to the reduction of fruit waste by proposing a method to detect fruit freshness or spoilage before it occurs.
2. A novel framework was developed that harnesses the capabilities of IoT sensors and machine learning techniques to effectively evaluate and monitor banana ripeness, offering a practical solution to the identified problem.
3. The successful integration of two key sensors, namely SHT40 and SGP30, alongside an ESP32 Development board, is demonstrated as the core component of the proposed system. These sensors play a crucial role in capturing emissions from fruits and facilitating the assessment of their ripeness and overall quality.
4. Through extensive experimentation employing various machine learning algorithms, the research achieved a high level of accuracy in classifying sensor data over a one-week period. Notably, the Cat-Boost classifier stands out for its exceptional accuracy in evaluating the collected data.

2. Literature review

The Internet of Things (IoT) is a system that integrates physical objects and communication technologies to create a networked world where devices can communicate with each other. Technologies such as Wi-Fi, Bluetooth, radio-frequency identification (RFID), and Wireless Sensor Networks (WSNs) enable interconnectivity. IoT goes beyond traditional web browsing and allows remote monitoring and control of physical environments [11]. The integration of modern technologies, such as the Internet of Things (IoT) and Machine Learning (ML), offers the potential to revolutionize the assessment of fruit ripeness. The IoT's ability to collect real-time data and ML's capacity to analyse complex datasets make it possible to develop automated, data-driven solutions for fruit ripeness detection. Despite numerous studies utilizing advanced technologies for ripeness detection, there is limited research examining the efficacy of these technologies in evaluating fruit ripeness and spoilage.

The emergence of IoT, Machine learning, and Artificial Neural Networks (ANN) impacted the monitoring and tracking of cold chain system in a radical way. Several studies focussed on harnessing the advantages of IoT in monitoring fruit quality, and ripeness. Some of the studies are provided here in the Table 1

Several researchers have used deep learning with image processing to detect fruit quality, while a limited number of others have turned to machine learning for the same objective. Table 2 provides some more investigations of fruit ripeness and quality assessment.

Despite much research in fruit ripeness and quality assessment, a few limitations are observed and are given below.

- The application of affordable sensors may be constrained in terms of their suitability for industrial purposes.

Table 1

IoT based investigations for fruit ripeness and quality.

Case study	Objectives & Outcomes	References
IoT based Intelligent Cold chain tracking system	Maintains product quality by tracking temperature humidity and location in real time. Alerts if any issues are detected.	[12]
Real-time evaluation of fruit and vegetable quality using IoT	An infrared camera, Arduino processor, and a Wi-Fi module were used to assess fruit ripeness on the plants	[13]
Fruit ripeness using IoT	Assesses apple ripeness using Arduino boards, and monitors banana ripening in real-time with WSNs and Xbee sensor nodes	[14,15]
Training an Artificial Neural Network (ANN) to classify fruit condition	Extracting temperature, humidity, ethylene and CO ₂ by a gas sensor to train an ANN. It resulted in an efficient diagnosis of fruit conditions and the importance of ethylene in fruit ripening.	[16]
Temperature management in the logistics of perishable items with the help of RFID and WSN	Proposed the potential advantages of combining RFID and WSN technologies in refrigerated storage to improve temperature regulation and energy efficiency.	[17]
Warehousing system with IoT sensors, microcontrollers and cloud connectivity	Enabled real-time monitoring and automated control of environmental parameters. It also provided web and mobile interfaces for data visualization and provided easy control.	[18]
Low-cost portable electronic nose with MQ2 and DHT11 sensors	Predicts fruit freshness and expiry date by analysing gas sensor values from bananas with 97.05 % accuracy. This has limitations in industrial applications	[19]
zNose-based device to assess ripening stages of bananas	Assessing ripening stages of bananas by evaluating their quality attributes, including respiratory quotient, total soluble solids, moisture content, and firmness by applying Principal Component Analysis (PCA) Multiple Linear Regression models. This model yielded 80 % accuracy.	[20]
Electronic nose system to identify spoilage in Filipino cuisines	It employed MQ gas sensors and an ANN algorithm to identify spoilage in Filipino cuisines with tomato-based ingredients	[21]
Hybrid approach for mango ripeness classification	It employed image processing and Odour-sensing techniques and achieved a 94.69 % correct classification rate	[22]
IoT monitoring system for grain storage	It addresses grain storage challenges. It uses sensors and microcontrollers to monitor temperature, humidity and grain quality. It follows <i>First In, and First Out</i> to optimise warehouse management and enhance fair grain distribution for food security.	[23]
Pizza spoilage detection	Employed cost-effective sensors, including CO ₂ , VOC, ethanol, pH, and NIR.	[24]
FruitSense – Ripeness detection system	It investigated the post-harvest freshness of papaya and watermelon by measuring CO ₂ , water vapour emissions, and O ₂ absorption. It employed Wi-Fi devices in residential settings.	[25]

(continued on next page)

Table 1 (continued)

Case study	Objectives & Outcomes	References
IoT and machine learning-based model for predicting fruit freshness	An IoT and machine learning system-based model for predicting fruit freshness as a per cent.	[26]

Table 2

IoT and ML-based investigations for fruit ripeness and quality.

Case study	Objectives & Outcomes	References
Ripeness classification in cape gooseberries using machine learning and colour spaces	The results indicate that the L*a*b* space coupled with the SVM classifier outperformed the others. Although the new PCA approach improved the model performance, it also increased the complexity	[27,28]
Tomato classification based on their colour	This is a monitoring system employing IoT, image processing, and machine learning algorithms including K-NN, MLP Neural networks, and K-means to classify tomatoes.	[29]
Tomato classification with Raspberry Pi and a Camera	A cost-effective monitoring system with the help of machine learning, and image processing algorithms such as Support Vector Machine (SVM), and CIELab colour space respectively. It will assess tomato ripeness automatically to reduce loss in the agricultural industry.	[30]
Machine learning (ML) based fruit ripeness system	This model utilised K-NN, SVM and decision tree methods to classify avocado and ripeness levels. It is suitable only for packaging, not for retail or cold supply chain applications.	[31]
Machine learning-based model for monitoring fruit quality	ML digital twin for monitoring banana fruit quality during storage using real-time thermal imaging data	[32]
ML-based approach to evaluate fruit quality using gas sensors	This model combines real-time data with MQ3 and MQ7 sensors with Arduino Uno processor, and PCA and LSVM classifiers to classify fresh and rotten apples. This is limited to the classification of two classes	[33]

- The methods used by researchers to assess/monitor the quality of fruits and vegetables in retail trays, or the cold storage supply chain may not be suitable for industrial settings.

The proposed framework evaluates the performance of state-of-the-art sensors for measuring the gases emitted by fruits, temperature, and humidity. Subsequently, advanced machine-learning algorithms were applied to predict the ripeness of the fruit based on the collected data. This investigation will focus primarily on bananas, as they are widely consumed in India. Besides that, they have numerous potential applications for research, particularly in detecting ripeness and spoilage. The rapid ripening process of bananas makes them an ideal subject for studying ripeness indicators. Analysis of gases emitted during various ripening stages can provide valuable information that can be correlated with machine learning algorithms. This study aimed to develop machine learning-based methods for fruit ripeness and spoilage detection, to enhance food quality management and supply chain logistics in practical settings. The proposed framework demonstrates remarkable accuracy in monitoring the quality of fruits and vegetables in retail store trays or cold storage supply chains, presenting a practical and feasible solution.

3. Methodology

Fig. 1. shows the schematic layout of the proposed experimental arrangement with all the necessary components. They include IoT-based sensor node design, cloud, dataset development, and machine learning.

3.1. Sensor node design

The sensor node is the combination of the microcontroller and the sensors. It consists of an ESP32 microcontroller, SHT40 temperature and humidity sensor, and SGP30 gas sensor. These components are connected as shown in the schematic diagram in Fig. 2. As ESP32 can provide Wi-Fi connectivity, and powerful processing, the sensor node is ready for connectivity and processing complex data also. ESP32 operates as the central processing unit of the sensor node, functioning as a microcontroller with built-in Wi-Fi capabilities that enable wireless communication and data transmission. The SHT40 Temperature & Humidity Sensor is incorporated into the Sensor node and measures temperature and humidity precisely with the help of capacitive sensing. The SHT40 sensor is interfaced with the ESP32 microcontroller using an I2C communication protocol. The I2C communication protocol is employed to retrieve humidity and temperature data from the sensor and convert it into relative humidity and temperature readings using appropriate mathematical formulas.

The SGP30 Indoor Air Quality Sensor developed by Sensirion, was specifically designed for precise gas sensing in indoor environments. This compact, energy-efficient sensor incorporates multi-pixel gas sensor technology, enabling accurate measurements of Total Volatile Organic Compounds (TVOC) and Carbon Dioxide Equivalent (CO_{2eq}) levels, as well as H₂ and Ethanol levels, which are essential for evaluating indoor air quality. The SGP30 sensor's I2C interface and advanced signal processing algorithms deliver comprehensive insights into indoor air quality, empowering users to create healthier living environments through appropriate actions. The SGP30 gas sensor operates within the recommended temperature and humidity ranges, ensuring optimal performance and longevity. Connecting the SGP30 gas sensor with the ESP32 microcontroller via I2C protocol enabled gas detection and air quality monitoring. Sensor node firmware on the ESP32 is developed and deployed with the help of Arduino IDE.

In the investigation, a plastic container was employed to store bananas, and a sensor node was attached to measure the gases emitted during the ripening process, as shown in Fig. 3. Time-series data were gathered from the raw bananas to the point of spoilage using this sensor node.

3.2. ThingSpeak cloud and data collection

The ThingSpeak platform, which is known for its dependability, plays a crucial role in the design of sensor nodes. Utilizing ThingSpeak's real-time data visualization platform provides immediate insights into the fruit ripening conditions. ThingSpeak comprises channels, each of which offers storage space for data. The initial step involves the creation of a ThingSpeak channel that serves as a virtual platform for gathering and processing sensors or device-generated data. During the setup phase, it is necessary to create an account on the ThingSpeak Cloud and configure essential parameters, such as privacy settings and metadata. Additionally, unique API keys must be obtained to facilitate seamless communication between the sensor node and the platform. ThingSpeak offers various applications such as ThingTweet, ThingHTTP, and React to facilitate integration. Custom applications can also be developed using HTML, CSS, and JavaScript to tailor the solutions to specific IoT project requirements. Overall, this study considers ThingSpeak as a versatile platform for IoT development with a range of features and tools. ThingSpeak significantly enriched this research by providing a robust infrastructure for data management and analysis, contributing to the success and scientific significance of this research.

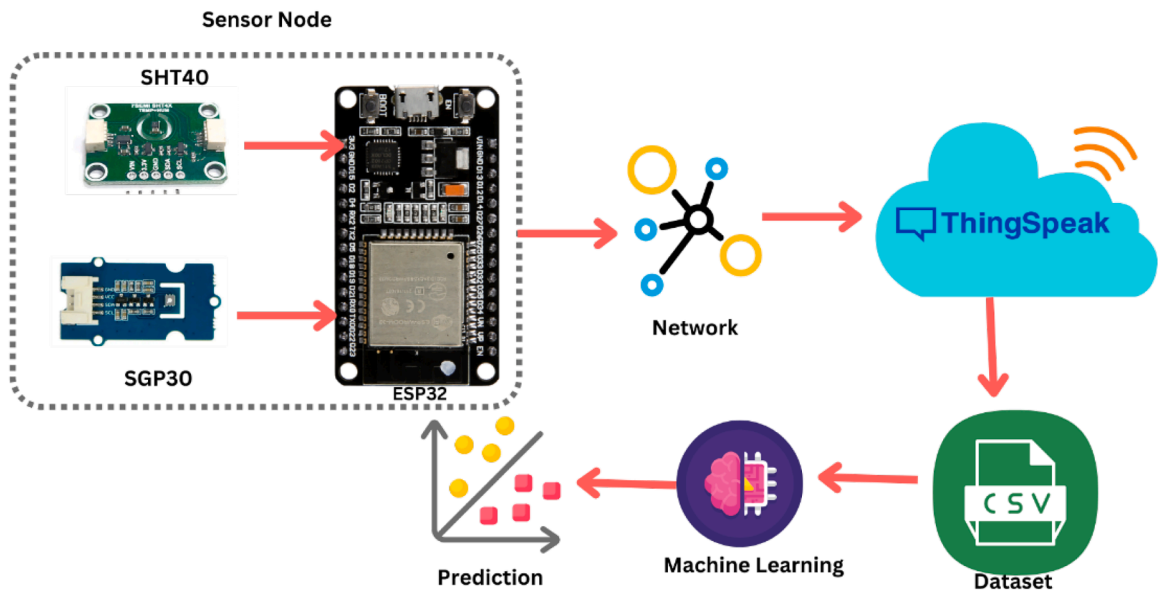


Fig. 1. Schematic layout of proposed prototype.

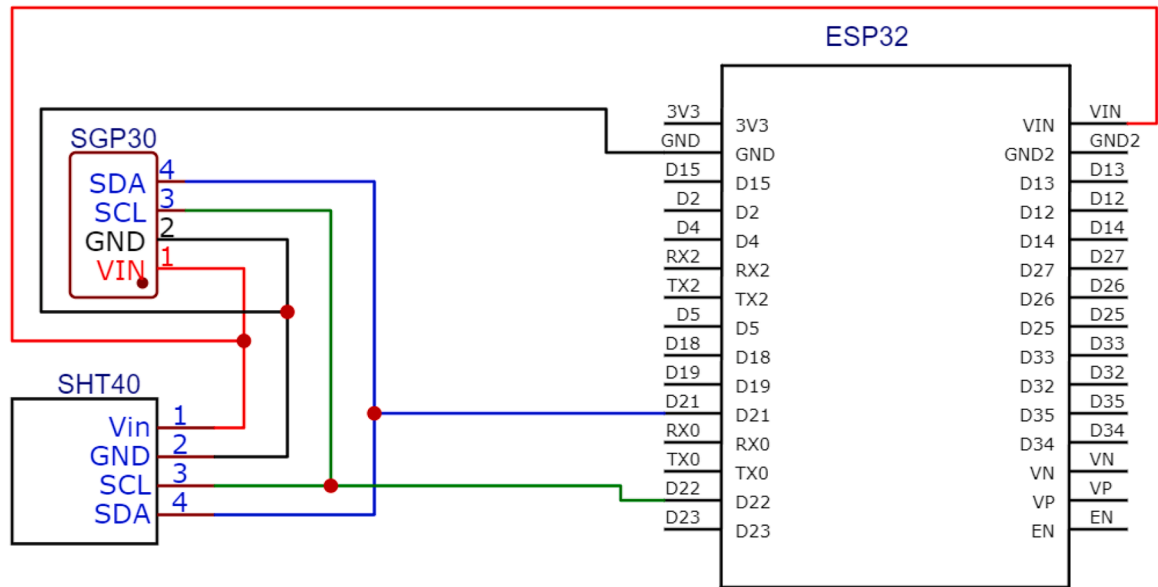


Fig. 2. Wiring diagram of ESP32, SGP30 sensor and SHT40 sensor.

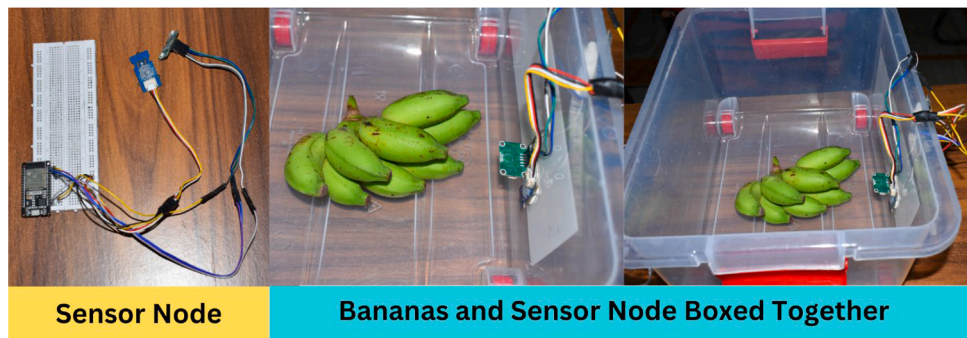


Fig. 3. Fruit Ripeness Detection experimental setup.

In the proposed prototype, the ESP32 microcontroller collects data from various sensors locally and transmits them wirelessly to the ThingSpeak IoT platform, where it is securely stored and visualized. Its user-friendly interface enabled efficient transmission of sensor data, including temperature, humidity, TVOC, CO₂, H₂, and ethanol measurements, from the ESP32 microcontroller to the cloud. A comprehensive dataset was developed through the collection of sensor data for 7 days. There are two methods to create this dataset. The ThingSpeak REST API in Python can be used to gather necessary data. The graphical representation of the total data collected during the banana ripeness process is given in Fig. 4.

The dataset can be accessed in the form of a CSV file from the ThingSpeak cloud platform, which provides users with all data collected from various sensors, including temperature, humidity, TVOC, CO₂, and other significant gas measurements. This extensive dataset is available for complete download and offers a valuable resource for researchers and professionals upon request. The variation of key parameters like temperature, humidity, and generation of CO₂, H₂, Ethanol and TVOC in the banana ripeness assessment is given in Fig. 4.

3.3. Data processing

This research incorporated rigorous data cleaning and feature engineering procedures to ensure the reliability and accuracy of the gathered data. During this phase, the data from the sensors and their relationship with the ripening process of bananas will be observed. These findings are essential for further exploration and analysis of experimental outcomes, ultimately contributing to a deeper understanding of the banana ripening process and the utility of sensor-based monitoring systems in agricultural contexts. Data pre-processing was performed using Python libraries such as NumPy and Pandas, focusing on several key tasks:

1. **Outlier Detection and Handling:** Outliers were identified and managed using the Interquartile Range (IQR) technique. This

approach involved calculating the IQR and applying it to filter out anomalous data points that could skew the analysis.

2. **Date Formatting:** Dates were standardized to ensure consistency across the dataset, facilitating accurate time-series analysis and comparisons.
3. **Null Value Management:** We systematically checked for and removed null values from the dataset to avoid any potential biases or inaccuracies in subsequent analyses.

These data-cleaning steps were crucial in transforming raw data into a well-structured format suitable for machine learning. By addressing outliers, ensuring consistent date formatting, and eliminating null values, we prepared the data for reliable and insightful analysis.

The dataset was carefully labelled with appropriate designations of fruit ripeness, encompassing the categories of not ripe, ripe, and spoiled. In the Data Analysis phase, we meticulously examined key variables, including temperature, humidity, H₂, Ethanol, CO₂, and TVOC. The data was categorized into three ripeness stages—Not ripe, Ripe, and Spoiled—based on the combination of visual inspection of camera images and TVOC values. Fig. 5 depicts the varying degrees of ripeness for bananas, including those that are not ripe, ripe, or spoiled. Data for each stage



Fig. 5. Banana ripeness state from raw to spoilage.

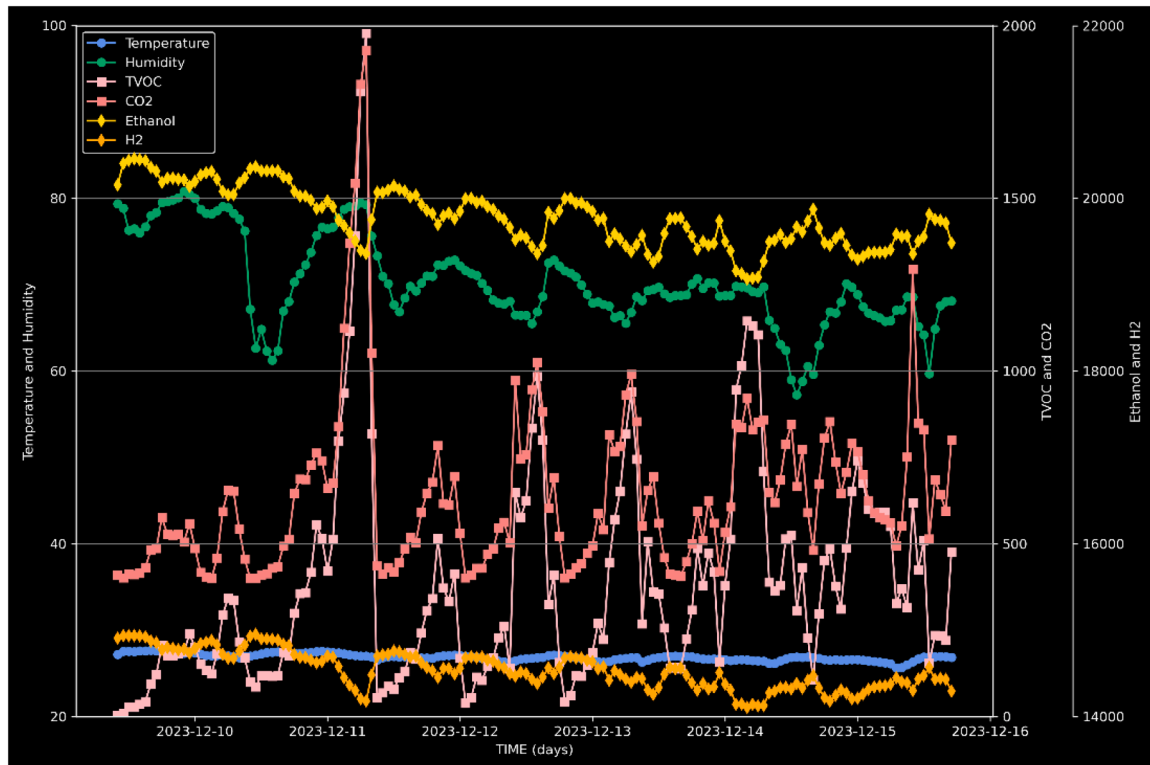


Fig. 4. Banana ripeness assessment data set in graphical form.

were collected using an experimental setup.

The criteria for these categories were as follows:

- **Not Ripe:** Characterized by a green to pale green colour, firm texture, and a thick, inflexible peel.
- **Ripe:** Marked by a bright yellow colour (with possible green tips), a softer yet still firm texture, and a thinner, more flexible peel. The presence of some brown spots may be observed.
- **Spoiled:** Identified by a yellow colour with extensive brown spots or a completely brown peel, a very soft and mushy texture, and significant changes in TVOC values.

These classifications enabled a structured analysis of the ripening process and supported the development of predictive models.

These criteria were consistently applied to both training and test sets for accurate classification. Following data cleaning, extensive data analysis was performed using appropriate statistical methods and techniques. The Key observations of this analysis can be observed from Fig. 7, and they include:

- **TVOC:** Increased in ripe and spoiled stages.
- **Ethanol:** Decreased in the spoiled stage.
- **H₂:** Decreased overall.
- **CO₂:** Increased in ripe and spoiled stages.

These findings are crucial for further exploration of experimental outcomes and contribute to a deeper understanding of the banana ripening process. They also highlight the utility of sensor-based monitoring systems in agricultural contexts. By examining these trends, a machine-learning model can be developed to predict fruit quality using this dataset.

3.4. Machine learning for fruit quality prediction

The categorization of fruits into varying stages of ripeness, including *not ripe*, *ripe*, and *spoiled*, was based on input features derived from the sensor data. The functionality of the machine learning model was assessed using pertinent evaluation metrics such as a classification report, confusion matrix, and accuracy scores. These metrics provide a comprehensive understanding of the model's predictive capabilities, precision in different ripeness categories, and overall accuracy in fruit ripeness prediction. The machine-learning approach demonstrated the potential for automation in fruit ripeness detection and highlighted the significance of employing interpretable models in agricultural and food supply chain applications, where transparency and explainability are crucial.

The research involved a rigorous machine-learning evaluation process to enable the device to predict banana ripeness stages based on sensor data. The dataset was divided into training and test sets. Various algorithms, including the Random Forest Classifier (RFC), Decision Tree Classifier (DTC), Support Vector Machines (SVM), AdaBoost Classifier (ADB), and CatBoost Classifier (CBC), were used to train the models on the training set. The trained models were then used to make predictions on the test set. After generating predictions, the performance of each model was evaluated by comparing the predicted ripeness stages with the actual labels from the test set. This evaluation provided valuable insights into the effectiveness and performance characteristics of each model, enabling informed decisions regarding their deployment in practical applications. To ensure a comprehensive evaluation of the machine learning models utilized in this study, detailed parameter settings and training procedures for each algorithm are essential. The configuration parameters for the ML algorithms considered are given below.

- **Random Forest Classifier (RFC):** The model was configured with 100 estimators and a random state of 42 to ensure reproducibility. The

RFC was trained on the normalized training dataset, and its performance was assessed using accuracy metrics on the test set.

- **Decision Tree Classifier (DTC):** It was set with a random state of 21 to facilitate consistent results. The training involved fitting the model to the normalized data, with performance evaluated based on test set predictions compared to actual labels.
- **K-Nearest Neighbors (KNN) algorithm:** The model was configured with 5 neighbours, a choice that balances sensitivity and specificity. The KNN model was trained on the normalized dataset, and its effectiveness was measured through predictions on the test set.
- **Support Vector Machine (SVM):** The model was configured with a random state of 21 to maintain consistency. The training involved fitting the model to the normalized training data, and performance was gauged based on its predictions on the test set.
- **CatBoost Classifier (CBC):** It underwent a rigorous hyperparameter optimization process using Optuna. Initial settings included a random seed of 21. Hyperparameters such as learning rate, depth, colsample_bylevel, min_data_in_leaf, and bootstrap_type were optimized. The learning rate ranged from 1e-3 to 0.1, depth from 1 to 10, colsample_bylevel from 0.05 to 1.0, and min_data_in_leaf from 1 to 100, with bootstrap_type selected from 'Bernoulli' and 'MVS'. The subsample parameter was optimized when bootstrap_type was not 'Bayesian'. The model was trained with the best-found hyperparameters, and accuracy was evaluated on the test set. **AdaBoost Classifier (ABC):** It was configured with 50 boosting iterations and a learning rate of 0.005. The model was trained on the normalized training data, and its performance was evaluated through predictions on the test set.

The comparison of accuracy in the applied models is given bellow in Table 3, and CatBoost algorithms proves its advantage among all the other algorithms.

4. Results and discussion

Data analysis was performed using the appropriate statistical methods and techniques. Upon analysing the primary characteristics of the dataset, several noteworthy relationships were discovered. Before that, it is better to review the physiological relationship between parameters considered in determining banana ripeness. The physiological relationship between the parameters measured—such as temperature, humidity, TVOC, ethanol, and other volatile compounds—and the banana ripening process is crucial for understanding the data obtained and its relevance.

- **Temperature:** As bananas ripen, their metabolic rate increases, leading to higher respiration and ethylene production, which in turn affects the temperature of the fruit. Studies have shown that temperature significantly impacts the ripening speed and quality of bananas. For instance, elevated temperatures can accelerate ripening, leading to faster colour change and texture softening.
- **Humidity:** Humidity levels influence the rate of moisture loss from bananas, affecting their texture and shelf life. High humidity can slow down the ripening process by reducing the rate of moisture loss, while low humidity can lead to premature dehydration and spoilage [34].

Table 3
The comparison of accuracy among the applied algorithms.

	RFC	DTC	KNN	SVM	ADB	CBC
Not Ripe	0.8932	0.9024	0.9291	0.8932	0.8886	0.9939
Ripe	0.9073	0.8879	0.923	0.9594	0.9169	0.9908
Spoiled	0.8608	0.8615	0.8573	0	0	0.9755

- Total Volatile Organic Compounds (TVOC) and Ethanol: TVOC and ethanol levels increase as bananas ripen due to the breakdown of starches and sugars into these volatile compounds [35].
- Ethanol, in particular, is a byproduct of the fermentation process during ripening. Elevated levels of ethanol are often associated with over-ripening and spoilage. TVOC levels are used to monitor the ripening progress and detect spoilage, as they reflect changes in the fruit's metabolic activities.

Understanding these physiological relationships helps in justifying the relevance of the measured parameters. By correlating these parameters with the ripeness stages, we can accurately classify the bananas' ripeness and improve storage and handling practices. This approach aligns with findings from various studies that emphasize the importance of these parameters in fruit ripeness assessment and quality control. For instance, Fig. 6 illustrates that temperature typically rises as bananas progress from their raw state to their ripened state and then drops as they transition to spoiled. Fig. 7. shows variations of H₂, Ethanol, TVOC and CO₂ in the fruit ripening process. Among the correlations identified,

the most significant was the positive relationship between Hydrogen (H₂) and Ethanol, with an impressive coefficient of 0.92. The strong positive correlation between H₂ and Ethanol highlights their mutual influence and underscores their importance in predicting banana ripeness. The other one is the correlation between TVOC and CO₂, there is a strong positive correlation of 0.76 between Total Volatile Organic Compounds (TVOC) and Carbon Dioxide (CO₂). This indicates that an increase in the TVOC concentration is accompanied by a corresponding increase in the CO₂ concentration, and vice versa. Furthermore, we found a robust negative correlation of -0.73 between TVOC and Ethanol. This suggests a negative relationship, where elevated TVOC concentrations are associated with decreased ethanol concentrations, and vice versa. The presence of both positive and negative correlations provides valuable insights into the complex interplay of features essential for accurate banana ripeness prediction. These findings enhance our understanding of the underlying mechanisms and improve the precision of the predictive model.

In evaluating the performance of various classifiers on a dataset containing labelled instances that represented different levels of

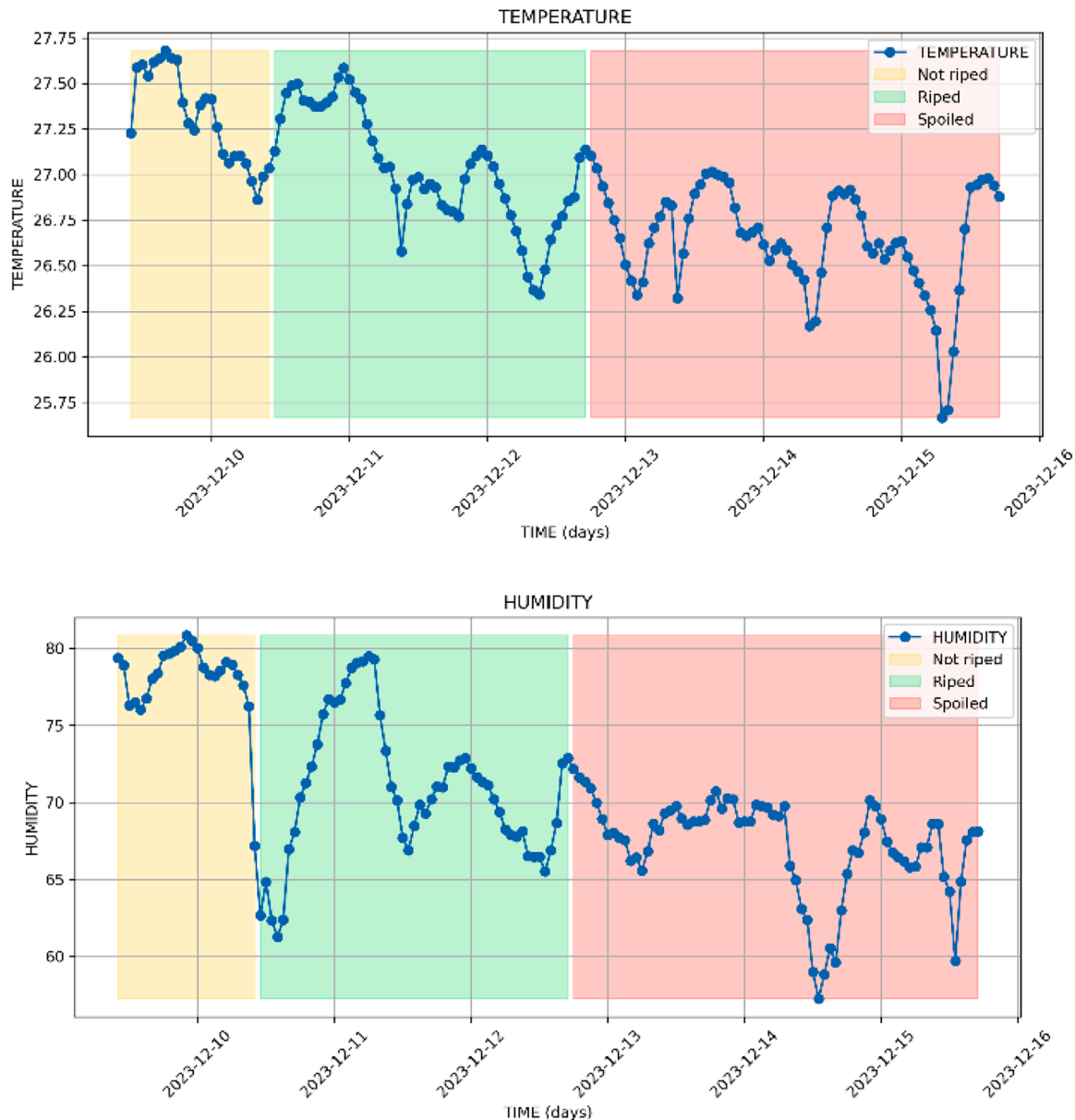


Fig. 6. Temperature Variations during the ripening process.

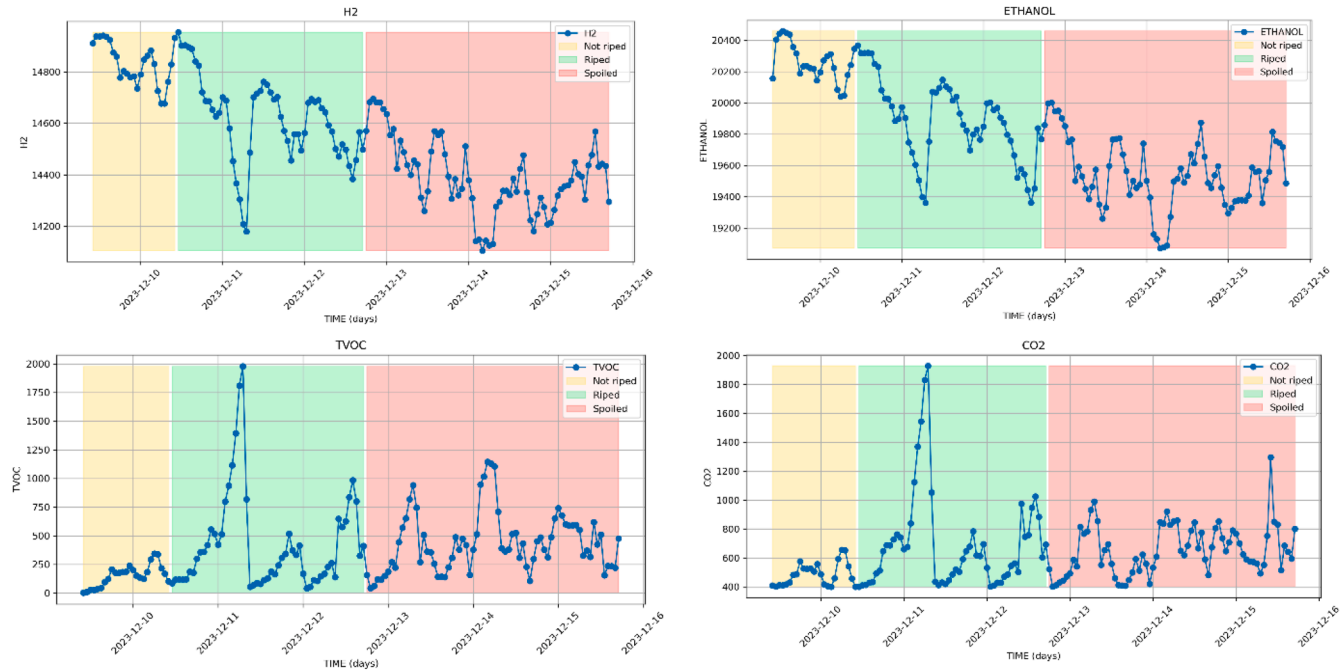


Fig. 7. H₂, Ethanol, TVOC and CO₂ levels in ripening process.

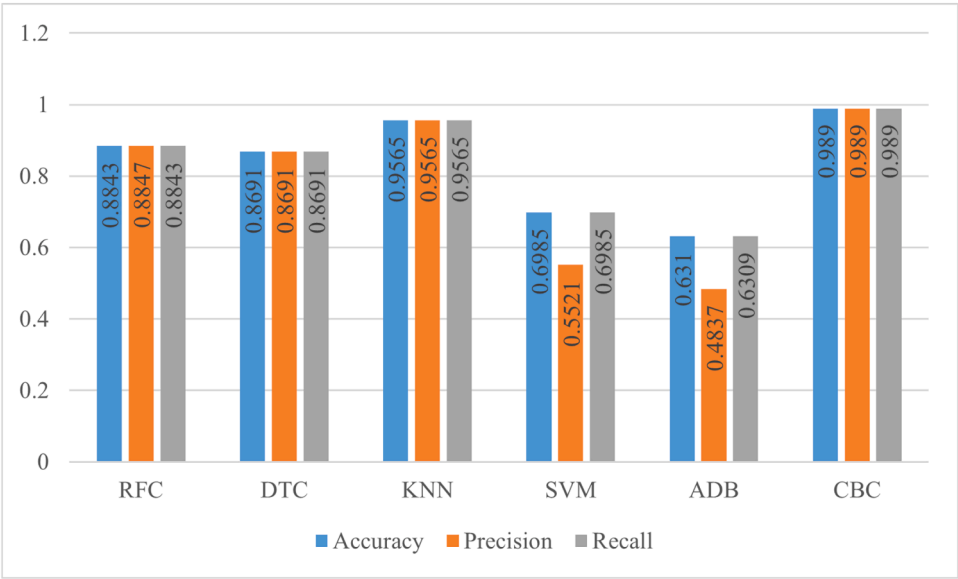


Fig. 8. Performance metrics for the considered ML algorithms.

ripeness, a range of performance metrics were observed. Fig. 8 provides a comparative analysis showcasing the performance metrics of different ML algorithms considered in this research. It can be observed from the table that, the CatBoost classifier outperformed all the other algorithms in providing better accuracy, precision and recall. The CatBoost model showed high class-specific accuracies: For Not ripe: 99.65 % accuracy, for Ripe: 98.90 % accuracy, and for Spoiled: 99.25 % accuracy. These results indicate that the model performs well in identifying the ripeness stages, with slightly lower accuracy for "Ripe" fruits.

This exceptional performance underscores its reliability and consistency in accurately categorizing fruits based on their ripeness. The outstanding results of the CatBoost classifier emphasize the critical importance of selecting an appropriate classifier based on the specific dataset characteristics and task requirements. The analysis is presented

in Fig. 8 showcases the advantage of CatBoost classifier among all the other algorithms. It presents the performance of all the algorithms considered and the possible reasons reduced performance. Our model compares favourably with earlier research in terms of accuracy, particularly in the context of monitoring fruit and vegetable quality in retail stores and cold-storage supply chains. Our framework demonstrated its suitability for specific applications, offering a reliable solution for quality assessment in these environments.

This research presented a proof-of-concept framework that combined IoT and ML technologies to increase the precision and efficiency of fruit ripeness evaluation. The experimental setup allowed for the real-time monitoring of environmental factors and banana ripening stages, leading to more efficient supply chain management. The comparative analysis with the existing research on fruit ripeness is shown in the

Table 4

Comparative analysis of existing and proposed frameworks.

Works	Application	Sensors used	Algorithm used	Accuracy	Limitations	Classes
[36]	Predicting the expiry date of bananas	MQ2 and DHT11	Logistic regression, DTC, SVC, KNN	97.05 %	Low Cost but not for Industrial use	2
[37]	Classification of banana fruit maturity using zNose	zNose Model 4200 Ultra-Fast GC Analyzer	PCA + Multiple Linear Regression	R2 values of 0.93, 0.94, 0.96 and 0.95	Useful for Laboratory Testing	2
[38]	Detecting spoilage in tomato-based Filipino cuisines	MQ137, MQ3, MQ7, MQ138, MQ5, MQ136, MQ135, DHT11 AM2302	Multi-layered Feedforward Artificial Neural Network with backpropagation	96.15 %	Useful for recipes	2
[39]	Classification of Mango ripeness	Camera, Gas Sensor Array	Image Processing + SVM	94.69 %	Useful for sorting and Automation	2
Proposed framework	Detection of Banana ripeness using ESP32, SGP30, and SHT30 sensors	SGP30 Gas sensor, SHT30	RFC, DTC, KNN, SVM, ADB, CBC	98.90 %	Useful for Retail and Cold storage supply chain	3

Table 4. It clearly emphasizes that the proposed model has achieved higher accuracy than the existing research and can be useful in transforming fruit ripeness and spoilage issues with much effectiveness.

5. Conclusion

In conclusion, the results of our study demonstrate the promising potential of integrating Internet of Things (IoT) and machine learning (ML) technologies in automating fruit ripeness assessment. Our proposed framework, which employed various ML algorithms, found that the CatBoost Classifier achieves high accuracy in predicting banana ripeness stages. This is high when compared to other studies focused on fruit ripeness. This superiority highlighted the dependability and efficiency of our model for advancing the field of fruit quality assessment. The scope of future research in this area is vast, with opportunities for exploration and improvement. The potential of the framework to assess ripeness stages in various fruits beyond bananas is worth investigating. Moreover, optimizing the algorithm and dataset can lead to increased accuracy and more versatile applications for fruit ripeness evaluation. This research lays the foundation for continued advancements in the use of IoT and ML technologies for automating fruit ripeness assessment.

CRedit authorship contribution statement

Rajini M: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Data curation, Conceptualization. **Persis Voola:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration.

Declaration of competing interest

The authors of this paper declare that they have no financial interests or personal relationships that could be perceived to have influenced the work presented in this study.

Data availability

Data will be made available on request.

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